BTS-III(S)-11-21-0916 Reg. No. | | | | | | | | |

C

B.Tech. Degree III Semester Supplementary Examination November 2021

SE 15-1303 ENGINEERING FLUID MECHANICS AND MACHINERY

(*2015 Scheme*)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) State Newton's law of viscosity. What is the effect of temperature on viscosity?

(b) Define capillarity. Derive an expression for capillary rise of a liquid.

(c) Differentiate Newtonian and non-Newtonian fluids.

(d) Explain the terms: path line, stream line, streak line and stream tube.

(e) What are the conditions of equilibrium of a floating body?

(f) Explain the terms meta-centre and meta-centric height.

(g) Explain the term hydraulic gradient.

(h) Obtain an expression for head loss due to sudden expansion.

(i) Define the terms speed ratio and specific speed of turbines.

(j) What is meant by operating characteristic curves of a pump?

PART B

$(4 \times 10 = 40)$

II. (a) Explain briefly about the commonly used pressure measuring instruments. (5)

(b) The right limb of a simple U-tube manometer containing mercury is open to atmosphere while the left limb is connected to a pipe in which a fluid of specific gravity 0.9 is flowing. The Centre of the pipe is 12 cm below the level of mercury in the right limb. Find the pressure of fluid in the pipe if the difference of mercury level in the two limbs is 20 cm. (5)

OR

III. (a) Derive an equation for the depth of centre of pressure from free surface of a vertical submerged plane surface by the static liquid. (5)

(b) Determine the total pressure on a circular plate of 1.5 m diameter which is placed vertically in water in such a way that the centre of the plate is 3 m below the free surface of water. Also find the position of centre of pressure. (5)

IV. Define the equation of continuity. Derive an equation for the continuity for a three dimensional flow. (10)

OR

V. (a) Differentiate between stream function and velocity potential function. (5)

(b) A stream function is given by $\psi = 2x - 5y$. Calculate the velocity components and also magnitude and direction of the resultant velocity at any point. (5)

VI. (a) What is boundary layer? Why does it increase with distance from upstream edge? (5)

(b) Derive Darcy- Weisbach equation. (5)

OR

VII. Differentiate between aggregative and particulate fluidization. (5)

Explain briefly pipe roughness and friction factor. (5)

(P.T.O)

VIII. Describe briefly the function of various main components of Pelton turbine with neat sketches. (10)

OR

IX. The diameters of an impellers of a centrifugal pump at inlet and outlet are 30 cm and 60 cm respectively. Determine the minimum starting speed of the pump if it works against a head of 30 m. (10)

***